

How does one choose a cloud solution provider? Would a private cloud or a public cloud be better, and how is security on the cloud handled? These are just a few of the many questions that cloud an IT manager's mind when evaluating cloud-based solutions. **Diksha P Gupta** from **Open Source For You** spoke to **Rahulkrishna Gupta, technical lead for Suse & Novell, India; Ashok Shenoy, regional manager, Datacenter and Cloud, from Cisco India & SAARC; and Rajesh Awasthi, director, Telecom & Cloud Service, NETAPP** to get some answers for you. Read on...

**Should an IT manager who's concerned about security go for a private cloud within the premises or opt for a vendor provided cloud?**

IT managers concerned about security should go for a private cloud. By deploying a private cloud, IT managers can meet the needs of the business, while lowering costs. And they can achieve this without losing control of their data, or opening their enterprise up to security risks. Private clouds allow IT managers to respond to the needs of the business more rapidly using their enterprise's infrastructure, security measures, and compliance protocols. More and more enterprises are making a private cloud implementation in their data centre a priority.

Use of the public cloud means that more of the organisation's data will travel along the public network and thus outside the direct control of the IT function. This is a major reason why CIOs would want to shift to a private cloud, which provides tightened security, increased business agility, improved data centre economics, and enables consistent cloud experiences for end users. Increased security controls in a private cloud also give CIOs a better control over data and availability. Besides, private clouds enable CIOs to integrate different legacy systems onto a single framework.

An on-premise private cloud, where control solely rests with the IT manager of the company, is more secure than a vendor provided cloud. A vendor provided cloud has two possibilities: either it is a private cloud with an off-premise set-up where the control lies with the vendor, or a public cloud where a particular server is shared. Looking at these options, it becomes evident why an on-premise cloud offering, where the server is also present within the company and the control lies with the IT manager, is more secure.

**What are the technical challenges involved in shifting from the typical infrastructure to one that is cloud-based?**

The primary technical challenges with such a shift are: converting physical workload(s) into cloud compatible workload(s), moving the data from normal infrastructure to cloud-based infrastructure, the need for adequate network bandwidth for accessing cloud-based infrastructure, and securing the data. Some of these challenges are not too applicable when dealing with a private cloud infrastructure since existing enterprise infrastructure, security measures and compliance protocols can be utilised:

It is imperative to get as many specifics as possible about the cloud provider's IDS, IPS, firewall, and other security appliances and technologies that have been deployed. The service provider's infrastructure should be protected as much as possible from threats such as Denial of Service (DoS) attacks, which can degrade the performance of a cloud service. If possible, perform an inspection of the cloud service provider's location where data will reside, in person. Finally, organisations should begin with a cloud service that doesn't require it to store critical or highly confidential data offsite, evaluate performance and security for a while, and then proceed to the next step, carefully.

Shifting from normal to cloud-based infrastructure is an expensive process. However, it is a one-time investment. The benefits associated with the cloud deployment far outweigh any growing pains that are experienced through the transition. The shift eventually delivers efficiencies and a tremendous amount of cost savings.